

1. Find the MLE of θ where the random samples are $X_1, \dots, X_n$ following one of the two pdfs:
 $f(x|\theta) = 1, 0 < x < 1$ else 0 when $\theta = 0$; or
 $f(x|\theta) = 1/(2\sqrt{x})$ when $0 < x < 1$ else 0 when $\theta = 1$.
2. Let $X_1, \dots, X_n$ random samples follow the exponential distribution with the parameter θ .
 - i. Find the MME of θ
 - ii. Find the bias of MME(θ) and then find the unbiased estimator of θ
3. Let $X_1, \dots, X_n$ random samples follow the truncated exponential distribution with the parameter θ having the following pdf:
 $f(x) = e^{-(x-\theta)}, x \geq \theta$ else 0.
 - i. Find the MLE of θ
 - ii. By using the Neyman-Fisher Factorization criteria, find a sufficient statistic of θ .
4. Let $X_1, \dots, X_n$ random samples follow the Normal distribution $N(0, \sigma^2)$. Given that

$$\sigma_{\text{hat}}^2 = 1/n * [\sum x_i^2]$$

- i. Examine if σ_{hat}^2 is an efficient estimator of σ^2
- ii. Find variance of σ^2

5. Estimate mean when standard deviation is given. Let there be n households and $X_1, \dots, X_n$ random samples denote time taken by a household to receive water supply follow some distribution. Then find the number of households required to be 95% sure that the sample mean is 15 mins within the population mean given that standard deviation σ is 35 mins.